

Fetal_Assessment_Project

08/08/2026 13:56:48

Thinking ● Inspected notebook cells and analyzed cough signal...



Thinking

- **Inspected notebook cells and analyzed cough signal exploration sections**

✓ Worked for 42s

Yes. I checked the **uploaded** `03_artifact_dataset_exploration(1).ipynb` **itself**, not just the earlier plan. The notebook currently contains the intended 13-stage story, including the newer frequency-band analysis cell.

`03_artifact_dataset_exploration(1).ipynb`

Below is the **study note I would save**. I'll explain what each cell is doing, what its output means, and-most importantly-how it contributes to your fetal-movement/artifact-rejection project.

`03_artifact_dataset_exploration.ipynb`

Complete Cell-by-Cell Understanding

CELL 1 - Imports

What this cell does

It imports the libraries required throughout the notebook:

- `Path` → handles dataset paths.
- `json` → reads `DataAnnotation.json`.
- `numpy` → numerical/signal calculations.
- `pandas` → tables and CSV data.
- `matplotlib` → plotting.
- `seaborn` → additional visualization.
- `scipy.fft` → FFT/frequency-domain analysis.
- `display` → clean notebook tables.

The notebook also sets display options and plotting style.

```
03_artifact_dataset_exploration(1).ipynb
```

Why it matters

This is simply the **computational environment**.

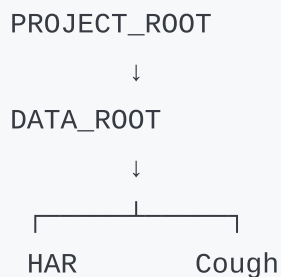
There is no dataset interpretation happening here.

What you should remember

"Cell 1 prepares the Python environment for dataset exploration, statistical analysis, visualization and frequency-domain analysis."

CELL 2 - Project Setup

The notebook defines:



Specifically:

```
data/raw/artifacts/har
data/raw/artifacts/cough_imu
```

The notebook is using project-relative paths rather than hardcoded dataset paths.

```
03_artifact_dataset_exploration(1).ipynb
```

Why this is important

This is connected to the Git workflow we discussed.

Your raw datasets can remain locally inside:

```
data/raw/
```

while `.gitignore` prevents them from being committed.

So the notebook itself can be pushed without pushing the actual datasets.

Research meaning

This establishes a **reproducible project structure**.

Instead of:

"My dataset is somewhere on my laptop."

you have:

"The artifact datasets are expected under the project's standardized raw-data directory."

That becomes important when we later create `dataset.py`.

CELL 3 - Dataset Discovery

This cell automatically searches both datasets.

For HAR:

```
har_csv_files = list(HAR_PATH.glob("*.csv"))
```

So it finds the CSV.

For Cough, it searches recursively for:

- accelerometer files
- gyroscope files
- magnetometer files
- WAV audio
- JSON annotations
- subject folders. `03_artifact_dataset_exploration(1).ipynb`

Your actual output

The notebook found approximately:

Dataset	What it found
HAR	1 CSV
Cough	13 subjects
Cough IMU	78 accelerometer
Cough	78 gyroscope
Cough	78 magnetometer
Cough	58 audio
Cough	1 JSON

The discovery output is visible in the notebook.

```
03_artifact_dataset_exploration(1).ipynb
```

Why 78 IMU files?

You have:

```
13 subjects × 3 trials = 39 recordings
```

and each recording has:

```
Accelerometer  
Gyroscope  
Magnetometer
```

Therefore:

```
39 × 2 = 78
```

for the two sensor modalities beyond the accelerometer count? More precisely, the notebook separately finds **78 files for each IMU modality**, so the dataset contains two copies/variants across the discovered trial structure. The important point for your story is that the dataset has extensive multimodal IMU coverage.

Why only 58 audio files?

Not every recording has both microphones/audio files. Your README already says some in-microphone recordings are missing.

So **58 audio files does not mean 58 subjects**.

CELL 4 - HAR Dataset Characterization

This is where we actually understand the HAR dataset.

The CSV contains:

```
ax
ay
az
gx
gy
gz
activity
```

So:

Accelerometer

```
X
Y
Z
```

Gyroscope

```
X
Y
Z
```

Label

```
activity
```

The dataset has:

```
15,980 samples
7 columns
6 activities
0 missing values
```

The notebook output confirms these values.

```
03_artifact_dataset_exploration(1).ipynb 03_artifact_dataset_exploration(1).ipynb
```

Activity IDs

The notebook later maps:

```
1 → Walking Upstairs
2 → Walking Downstairs
3 → Walking
4 → Sitting
5 → Standing
6 → Jogging
```

Why this dataset is useful

This dataset gives us **labelled human motion**.

That is extremely useful because your future fetal-monitoring sensor will not only observe fetal movement.

It will also observe:

```
mother walking
mother sitting
mother standing
mother changing posture
mother moving intensely
```

Those movements can contaminate the fetal signal.

So HAR gives us:

general maternal/body-motion artifact examples.

CELL 5 - HAR Signal Exploration

This cell takes a short portion of the HAR signal and plots the sensor channels.

The current code uses roughly the first 1000 samples and creates acceleration/gyroscope plots. `03_artifact_dataset_exploration(1).ipynb`

What are we looking at?

For acceleration:

```
ax
ay
az
```

we see how the body is moving over time.

For gyroscope:

```
gx
gy
gz
```

we see how the body is rotating.

Why this matters

Imagine:

Sitting

Signal:

```
_____
_____
_____
```

relatively stable.

Walking

Signal:

\~ \~ \~ \~ \~ \~ \~ \~

repetitive oscillation.

Jogging

Signal:

~~~ ~~~ ~~~ ~~~

generally stronger/faster motion.

This is exactly the kind of information we want to eventually distinguish from fetal movement.

## Key concept

**Time-domain signal = what the motion looks like over time.**

## CELL 6 - HAR Activity Distribution

This cell counts how many samples belong to each activity.

Your actual output is:

| Activity           | Samples |
|--------------------|---------|
| Walking Upstairs   | 2084    |
| Walking Downstairs | 2035    |
| Walking            | 2863    |
| Sitting            | 3619    |
| Standing           | 2585    |
| Jogging            | 2794    |

These values are directly present in the notebook output.

```
03_artifact_dataset_exploration(1).ipynb
```

## What does this tell us?

The dataset isn't completely balanced.

For example:

```
Sitting → 3619  
Walking Downstairs → 2035
```

So sitting has substantially more samples.

## Why does this matter for ML later?

If we train a classifier directly on this data, the model could become biased toward classes with more samples.

Therefore later, during artifact ML:

```
class distribution  
    ↓  
imbalance handling  
    ↓  
weighted loss / resampling / appropriate metrics
```

will matter.

## Research interpretation

This cell answers:

*"What artifact behaviours are actually represented, and how much data do we have for each?"*

---

## CELL 7 - HAR Frequency Analysis

This is one of the most important cells conceptually.

The notebook selects **walking** and applies FFT to the `ax` signal. It uses 2048 samples and currently assumes `fs = 50 Hz`. `03_artifact_dataset_exploration(1).ipynb`

The output was:

```
Estimated Dominant Frequency : 1.20 Hz
```

## What is frequency?

Suppose a signal repeatedly moves up/down.

In the time domain you see:



Frequency asks:

***How rapidly is this pattern repeating?***

If something repeats once every second:

```
1 cycle/second = 1 Hz
```

If it repeats ten times per second:

```
10 cycles/second = 10 Hz
```

## What is FFT?

FFT = **Fast Fourier Transform**.

It converts:

```
Time-domain signal
  ↓
  FFT
```



Frequency-domain signal

Instead of asking:

*"What is acceleration doing at time 3.5 seconds?"*

we ask:

*"How much signal energy exists around 1 Hz, 5 Hz, 10 Hz, etc.?"*

---

## What does the 1.20 Hz result mean?

If the **50-Hz sampling assumption is verified**, a dominant frequency around 1.2 Hz means that the selected walking signal has strong periodic activity around that frequency.

It does **not** mean:

*"Walking always happens at 1.2 Hz."*

It means:

*"In this selected segment, the strongest spectral component occurred around 1.2 Hz."*

This distinction is important when talking to your professor.

---

## Why is frequency analysis important for YOUR project?

Because your fetal movement signal and maternal artifacts can overlap in the time domain.

A maternal movement might look similar to fetal movement in amplitude.

But its **spectral structure, duration and multi-axis behaviour** may be different.

That gives us another feature family:



Time-domain  
+  
Frequency-domain  
+  
Sensor fusion

## Very important correction

The notebook currently uses:

```
fs = 50
```

as an **estimated** sampling rate, not a verified one.

```
03_artifact_dataset_exploration(1).ipynb
```

Therefore:

***Do not present 1.20 Hz as a scientifically finalized HAR frequency result yet.***

We need to verify the HAR dataset's true sampling frequency first.

## CELL 8 - Cough Dataset Characterization

Now the notebook changes from general motion to the much more relevant **chest-mounted multimodal dataset**.

It discovers:

```
13 subjects
3 trials/subject
Accelerometer
Gyroscope
Magnetometer
Audio
Synchronization files
JSON annotations
```

The actual notebook output confirms 13 subjects, 78 IMU files for each modality, 58 audio files and 13 synchronization files. `03_artifact_dataset_exploration(1).ipynb`

## Why this dataset is especially valuable

The sensor is:

### ***Chest mounted IMU***

according to the notebook. `03_artifact_dataset_exploration(1).ipynb`

That is much closer to your eventual wearable configuration than a generic activity dataset.

---

## CELL 9 - Cough Signal Exploration

This cell loads representative:

```
Accelerometer  
Gyroscope  
Magnetometer
```

files.

The accelerometer data contains:

```
epoc (ms)  
timestamp  
elapsed (s)  
x-axis (g)  
y-axis (g)  
z-axis (g)
```

The notebook shows, for example, elapsed times:

```
0.000  
0.010  
0.020  
0.029  
0.039 ...
```

and three acceleration axes. `03_artifact_dataset_exploration(1).ipynb`

## What does this tell us?

The dataset gives us **real continuous IMU waveforms** rather than only activity labels.

So now we can investigate:

```
amplitude
duration
periodicity
frequency
energy
multi-axis response
```

---

## Why the three sensors matter

### Accelerometer

Measures translational acceleration.

Useful for:

```
walking
body movement
cough-induced chest movement
posture changes
```

### Gyroscope

Measures rotational motion.

Useful when two events have similar acceleration but different body orientation/rotation.

### Magnetometer

Measures magnetic-field orientation.

Useful mainly for orientation/context.

## Audio

Provides an additional modality for:

```
cough  
speech  
laugh  
sneeze  
breathing
```

This is why the dataset is called **multimodal**.

---

## CELL 10 - Cough Event Analysis

This is actually a very important cell, but there is a subtle thing to understand.

The JSON contains annotation records with fields such as:

```
start_time  
end_time  
filename  
original_filename  
value  
segmentations
```

and event categories including:

```
Cough  
Groan  
Laugh  
Other Sound  
Sneeze  
Speech  
Speech (far)
```

The notebook's recursive inspection discovers these structures.

```
03_artifact_dataset_exploration(1).ipynb 03_artifact_dataset_exploration(1).ipynb
```

## Why this is important

The annotation isn't simply:



```
0 = no cough  
1 = cough
```

It contains **event information associated with audio recordings**.

The timestamps such as:

```
start_time  
end_time
```

mean that eventually we can connect:

```
Audio event  
  ↓  
Timestamp  
  ↓  
IMU timestamp  
  ↓  
IMU segment
```

That is extremely important for your future multimodal artifact modelling.

---

## One important limitation of the current Cell 10

The current cell is primarily an **annotation-structure discovery cell**.

It successfully discovers event names and JSON fields, but it does **not yet produce a scientifically clean count of actual cough events by category**.

The output currently includes many metadata values such as filenames, usernames and IDs. `03_artifact_dataset_exploration(1).ipynb`

So if you later want:

```
Cough = 127  
Speech = 300  
Laugh = 42  
...
```

we should build a proper parser based on the actual JSON schema.

I would **not call the current event-counting result final**.

# NEW FREQUENCY-BAND CELL - Very Important

Your notebook now contains the frequency-band framework:

```
0-1 Hz
1-2 Hz
2-5 Hz
5-10 Hz
10-20 Hz
20-30 Hz
>30 Hz
```

The exact bands are present in the notebook.

```
03_artifact_dataset_exploration(1).ipynb
```

This is where the notebook becomes much more relevant to your actual fetal-movement research.

## What is "frequency-band energy"?

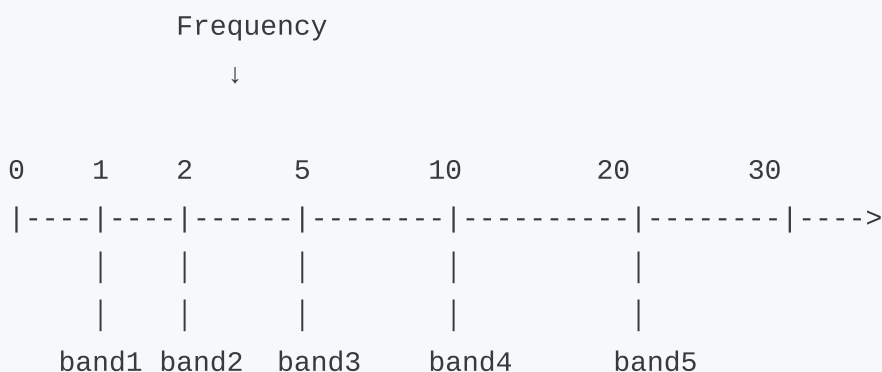
Suppose your signal contains:

```
1 Hz component
5 Hz component
15 Hz component
25 Hz component
```

FFT gives us the spectrum.

We then divide that spectrum into regions.

For example:



Then we calculate:

*How much of the signal's spectral energy lies inside each band?*

## Why energy instead of just dominant frequency?

Imagine two signals:

### Signal A

Strong 5 Hz component

### Signal B

Weak 5 Hz component

Both have:

Dominant frequency = 5 Hz

but their energy is completely different.

Therefore:

Dominant frequency  
+  
Band energy

is much more informative.

## Why these particular bands?

Your fetal-movement research already uses frequency-domain features based on:

1-2 Hz  
2-5 Hz  
5-10 Hz

10-20 Hz

20-30 Hz

including dominant frequency and spectral energy. fetal\_movements\_2024\_1 (1).pdf

And that work uses a **1-30 Hz band-pass filter** for accelerometer/fetal-movement sensing. fetal\_movements\_2024\_1 (1).pdf

So our artifact notebook adopting the same frequency windows creates a very useful bridge between:

Artifact datasets

↓

Fetal datasets

↓

Common frequency features

## But don't interpret the bands too literally

This is very important.

Do **not** say:

0-1 Hz = breathing

1-30 Hz = fetal movement

>30 Hz = artifact

as an established physical law.

Instead:

0-1 Hz

→ investigate slow/respiratory components

1-2 Hz

→ low-frequency movement

2-5 Hz

→ low-frequency movement

5-10 Hz

→ moderate-frequency movement

10-20 Hz

→ higher-frequency movement

20-30 Hz

→ high-frequency movement

>30 Hz

→ investigate high-frequency noise/artifact

These are **analysis bands**, not final class definitions.

## Why?

Because maternal artifacts can overlap with fetal-movement frequencies.

Your literature explicitly points out that maternal movements can create signals similar to true fetal movement and that filtering alone cannot completely eliminate these artifacts. [fetal movements\\_2024\\_2.pdf](#)

Another study found that maternal respiration concentrates in lower frequency regions, while fetal movement appears in shorter time ranges, and maternal laughing has a different spectrogram pattern. [journal.pone.0254560 \(1\).pdf](#)

That is exactly why **time + frequency** is stronger than frequency alone.

## What the frequency cell actually calculates

The function:

```
calculate_band_energy()
```

does approximately:

Raw signal

↓

Remove NaN

↓

Remove mean/DC component

↓

FFT

↓  
Power spectrum  
↓  
Divide into frequency bands  
↓  
Calculate energy  
↓  
Calculate energy percentage

The implementation calculates spectral power as:

```
power = |FFT|2
```

and then sums power inside each frequency band.

```
03_artifact_dataset_exploration(1).ipynb
```

## Output conceptually becomes:

| Frequency Band | Energy | Energy % |
|----------------|--------|----------|
| 0-1 Hz         | ...    | ...      |
| 1-2 Hz         | ...    | ...      |
| 2-5 Hz         | ...    | ...      |
| 5-10 Hz        | ...    | ...      |
| 10-20 Hz       | ...    | ...      |
| 20-30 Hz       | ...    | ...      |
| >30 Hz         | ...    | ...      |

## Why this will be VERY useful later

Suppose eventually we have:

### Fetal movement

High 2-5 Hz energy

Moderate 5-10 Hz

## Maternal walking

Strong 1-2 Hz periodic energy

## Maternal coughing

Short-duration burst  
Broad frequency distribution

## Maternal respiration

Strong low-frequency component

Then frequency-band energy becomes a feature.

Instead of the model seeing only:

raw acceleration

it sees:

```
mean
std
RMS
energy
duration

+

dominant frequency
energy_1_2Hz
energy_2_5Hz
energy_5_10Hz
energy_10_20Hz
energy_20_30Hz
```

That is a **much richer representation**.

---

## CELL 11 - Cross Dataset Comparison

This is the bridge between the two artifact datasets.

The notebook compares:

```
HAR
vs
Multimodal Cough
```

across:

- purpose
- subjects
- activities/events
- accelerometer
- gyroscope
- magnetometer
- audio
- chest mounting
- labels
- multimodal nature
- artifact-learning suitability. `03_artifact_dataset_exploration(1).ipynb`

### The important idea

The two datasets are **not duplicates**.

They provide different kinds of information.

### HAR

Primarily gives:



whole-body movement

## Cough

Gives:

```
chest movement
+
respiratory events
+
audio
+
IMU
```

So they complement one another.

## CELL 12 - Unified Artifact Library

This is probably the **most strategically important cell** in the notebook.

It converts the raw dataset classes into a conceptual artifact library:

| Artifact           | Source | Future Role               |
|--------------------|--------|---------------------------|
| Walking            | HAR    | Maternal motion           |
| Walking Upstairs   | HAR    | Maternal motion           |
| Walking Downstairs | HAR    | Maternal motion           |
| Standing           | HAR    | Posture                   |
| Sitting            | HAR    | Posture                   |
| Jogging            | HAR    | High-intensity motion     |
| Speech             | Cough  | Respiratory/body artifact |
| Laugh              | Cough  | Respiratory/body artifact |
| Deep Breath        | Cough  | Respiratory artifact      |

| Artifact   | Source | Future Role               |
|------------|--------|---------------------------|
| Cough      | Cough  | Respiratory artifact      |
| Groan      | Cough  | Respiratory/body artifact |
| Sneeze     | Cough  | Respiratory artifact      |
| Transition | Both   | Motion transition         |

The actual notebook creates this table. `03_artifact_dataset_exploration(1).ipynb`

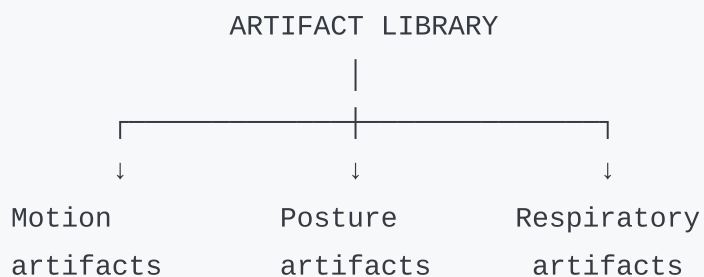
## Why this is more than just a table

This is essentially the beginning of your future software architecture.

Instead of thinking:

Dataset A  
Dataset B  
Dataset C

we start thinking:



Then later:

Raw maternal/fetal signal  
↓  
Artifact detection  
↓  
Artifact classification  
↓  
Remove / suppress artifact



Fetal movement analysis

## CELL 13 - Final Conclusions

This cell converts everything into the final research story.

The notebook concludes that:

### HAR contributes

Walking  
Sitting  
Standing  
Jogging  
Posture/motion changes

### Cough contributes

Cough  
Speech  
Laugh  
Deep breathing  
Other chest/body motion

Together they form an **initial artifact library** for the future artifact-rejection stage.

```
03_artifact_dataset_exploration(1).ipynb
```

And critically:

*These datasets **do not replace pregnancy-specific data**.*

The notebook explicitly states that they will eventually be complemented/replaced by data collected using your own STM32 maternal wearable.

```
03_artifact_dataset_exploration(1).ipynb
```

## The COMPLETE STORY of this notebook

If your professor asks:



## ***"Why did you analyze these two datasets?"***

You can explain it like this:

*"Our fetal datasets contain fetal movement signals, but the maternal body and sensor can introduce artifacts such as walking, posture changes, coughing, laughing, breathing and other body movements. Therefore, before developing the artifact-rejection module, we characterized publicly available IMU datasets representing these types of non-fetal motion."*

Then:

### **Dataset 1 - HAR**

```
Whole-body movement
    ↓
Walking
Standing
Sitting
Jogging
Stairs
    ↓
General maternal motion artifacts
```

### **Dataset 2 - Multimodal Cough**

```
Chest-mounted IMU
    +
Audio
    +
Annotations
    ↓
Cough
Speech
Laugh
Breathing
Sneeze
Body motion
```

↓  
Chest/respiratory artifacts

Then:

```
graph TD
    A[BOTH DATASETS] --> B[Artifact Library]
    B --> C[Time-domain characterization]
    C --> D[+]
    D --> E[Frequency-domain characterization]
    E --> F[+]
    F --> G[IMU features]
    G --> H[Future Artifact Model]
    H --> I[Artifact Rejection Module]
    I --> J[Fetal Movement Detection]
    J --> K[Fetal Behaviour Profiling]
```

## The frequency story you should remember

This is the part I'd especially save.

### Time domain asks:

*"What does the signal do over time?"*

### Frequency domain asks:

*"What rates of oscillation are present in the signal?"*

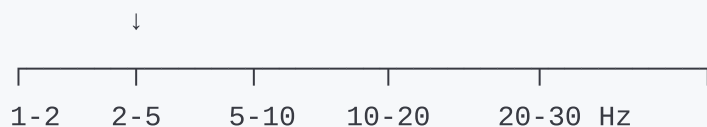
### FFT does:

Signal vs Time  
↓  
FFT

↓  
Signal vs Frequency

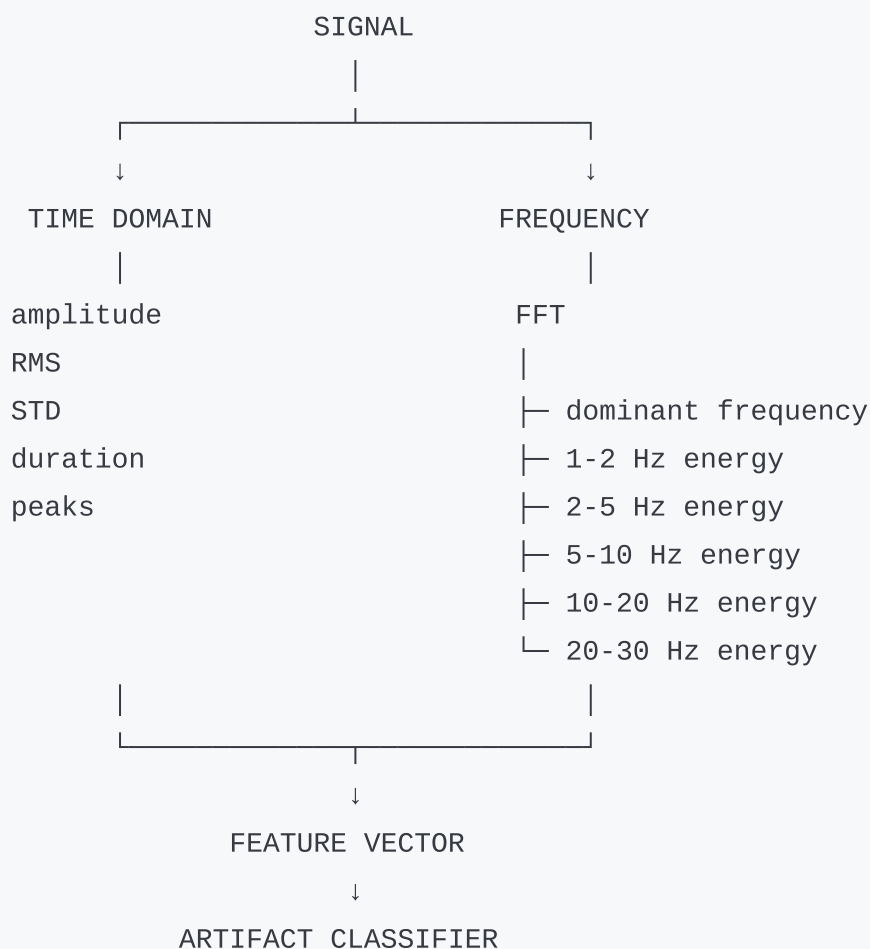
## Band-energy analysis does:

Frequency spectrum



↓  
Energy in each region

And eventually:



This is essentially the same philosophy used in the fetal-movement literature you have already been studying: dominant frequency plus spectral energy in **1-2, 2-5, 5-10, 10-20 and 20-30 Hz** bands. fetal\_movements\_2024\_1 (1).pdf

# One thing I would NOT forget before calling the notebook 100% final

There are **two technical caveats** in the current uploaded notebook:

1. **HAR** `fs = 50` **is currently an estimate**, so its absolute FFT frequency result should not yet be treated as verified. `03_artifact_dataset_exploration(1).ipynb`
2. The Cough JSON cell currently performs **schema/annotation discovery**, not a final scientifically validated event-count table. Its output contains metadata such as filenames and usernames in addition to event-related information.

`03_artifact_dataset_exploration(1).ipynb`

Everything else gives us a very good **dataset-characterization → artifact-library** story.

And this is the key transition for the project:

***The fetal notebooks tell us what fetal signals look like. The artifact notebook tells us what non-fetal maternal signals can look like. The next `dataset.py` layer will give all of them one common interface.***

That is exactly why this notebook comes **before** the unified signal-processing pipeline.

8/8/2026, 1:50:08 PM